

4 (a) (i) 0.050 m A1

(ii) $\omega = \frac{v_0}{x_0}$ C1

$$T = \frac{2\pi}{\omega}$$

$$0.42 = \frac{2\pi \times 0.050}{T} \quad \text{C1}$$

$$T = 0.75 \text{ s} \quad \text{A1}$$

(b) one point labelled P where ellipse crosses displacement axis B1

5 (a) When two or more waves of the same kind meet at a point at the same time B1